Experiment 11

```python
import pandas as pd

# ----------------------------------------
# 1. Create employee data
# ----------------------------------------

empdata = [[1, "Smith", 50000], [2, "Jones", 60000]]

# ----------------------------------------
# 2. Create a DataFrame from the data
# ----------------------------------------

df = pd.DataFrame(empdata)

print("Original DataFrame:")
print(df)

# ----------------------------------------
# 3. Assign column names
# ----------------------------------------

df.columns = ["Empd", "Name", "Salary"]

print("\nDataFrame with column names:")
print(df)

# ----------------------------------------
# 4. Display DataFrame information
# ----------------------------------------

print("\nDataFrame Information:")
df.info()

# ----------------------------------------
# 5. Get the column names
# ----------------------------------------

empCol = list(df.columns)

print("\nEmployee Column Names:")
print(empCol)


# ----------------------------------------
# 8. Convert DataFrame to NumPy array
# ----------------------------------------

emparray = df.values

print("\nEmployee Data as NumPy Array:")
print(emparray)


# ----------------------------------------
# 9. Convert NumPy array back to DataFrame
# ----------------------------------------

employee_DF = pd.DataFrame(emparray, columns=empCol)

print("\nNew Employee DataFrame:")
print(employee_DF)
```

✓ 0.2s

```
Original DataFrame:
   0      1      2
0  1  Smith  50000
1  2  Jones  60000

DataFrame with column names:
   Empd   Name  Salary
0     1  Smith   50000
1     2  Jones   60000

DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      str
 2   Salary  2 non-null      int64
dtypes: int64(2), str(1)
memory usage: 180.0 bytes

Employee Column Names:
['Empd', 'Name', 'Salary']

Employee Data as NumPy Array:
[[1 'Smith' 50000]
 [2 'Jones' 60000]]

New Employee DataFrame:
  Empd   Name Salary
0    1  Smith  50000
1    2  Jones  60000
```